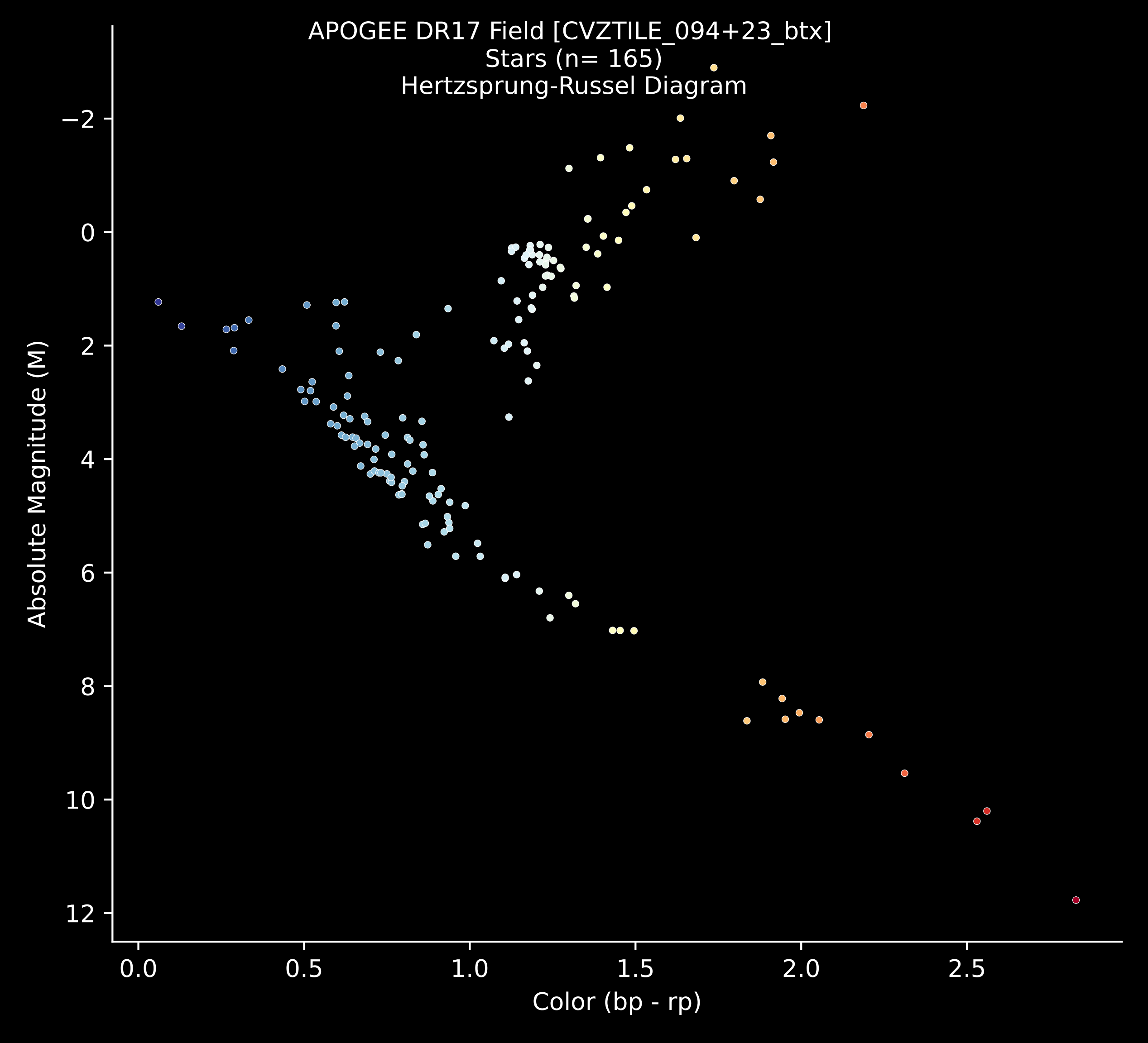
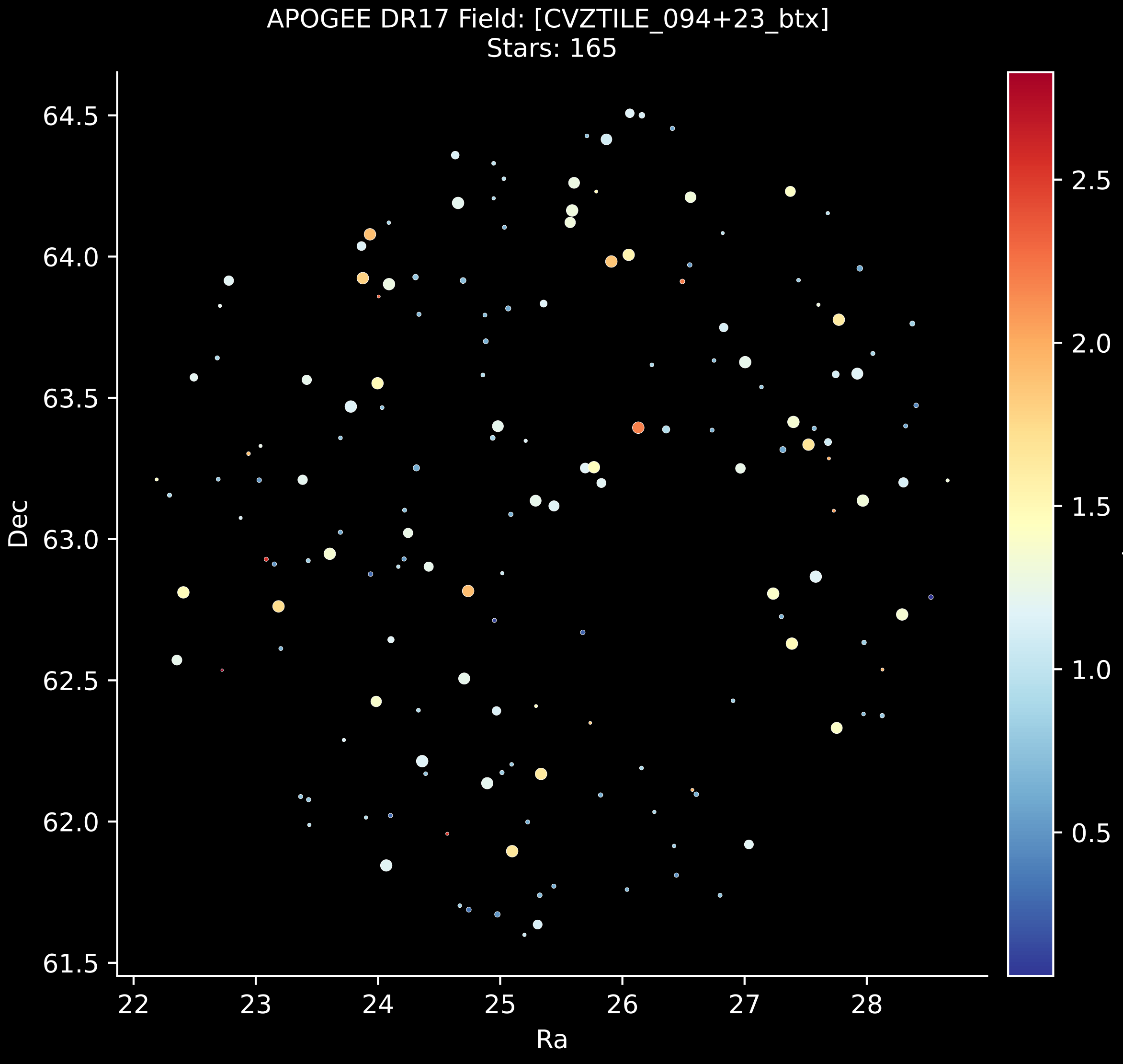
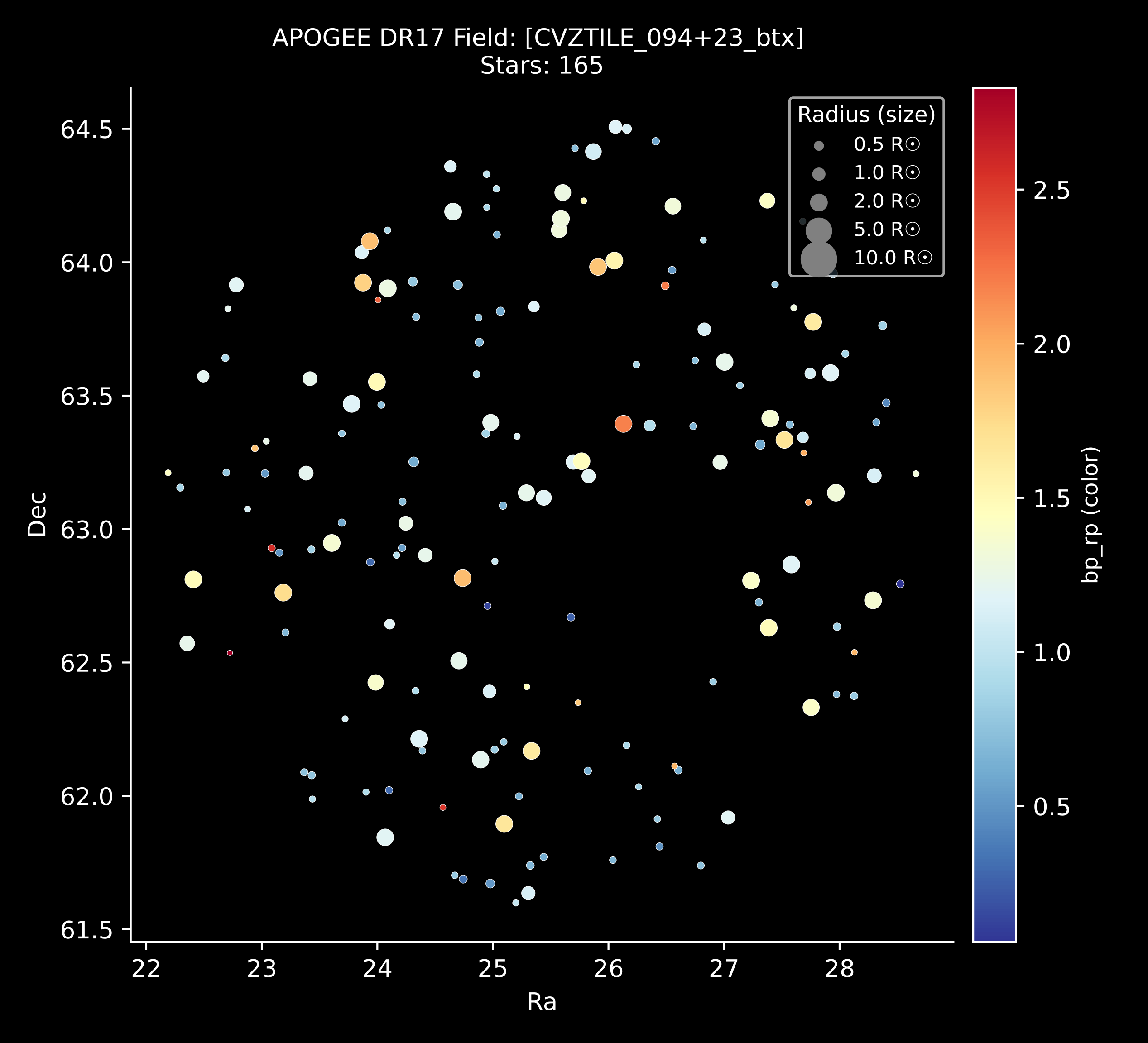
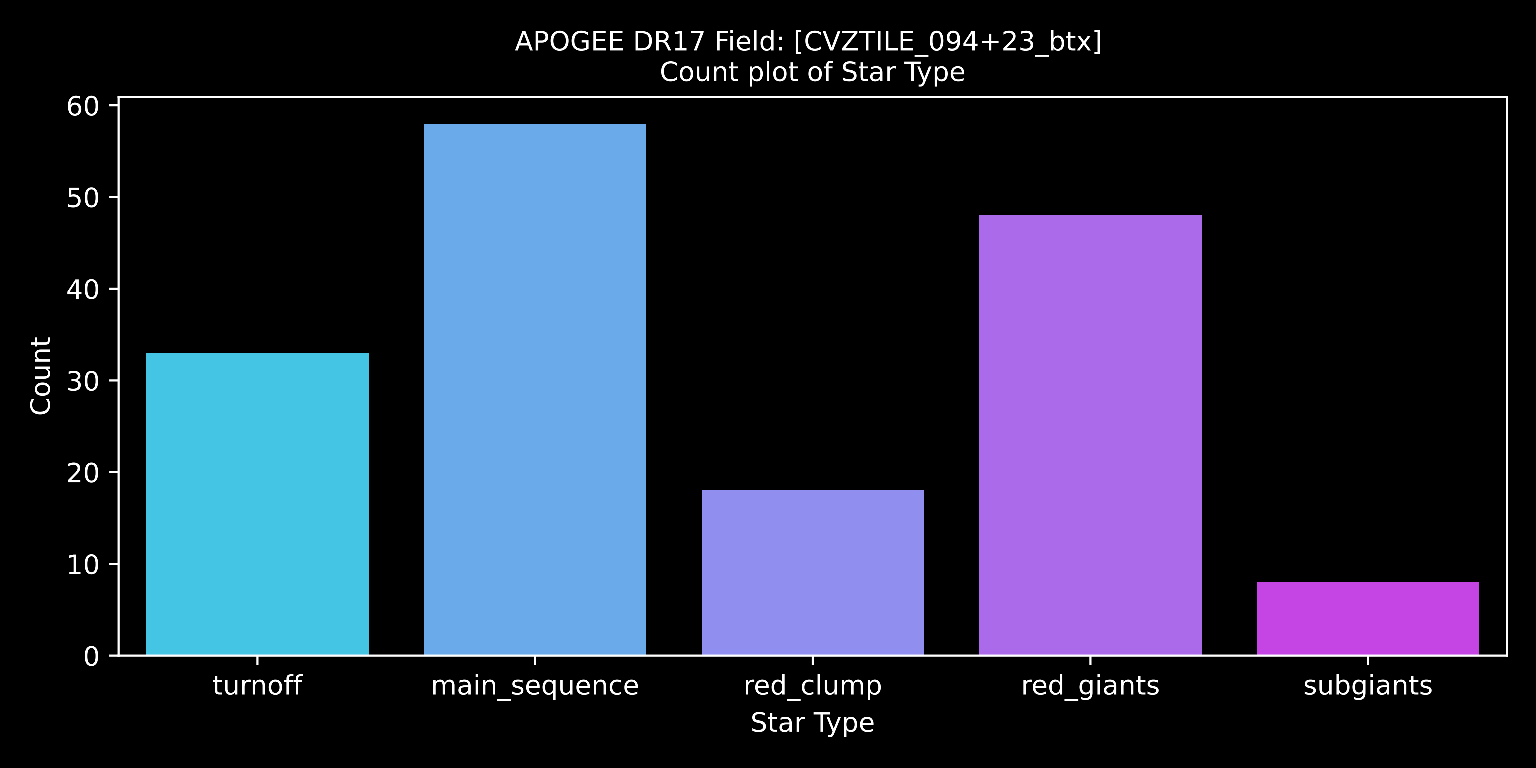
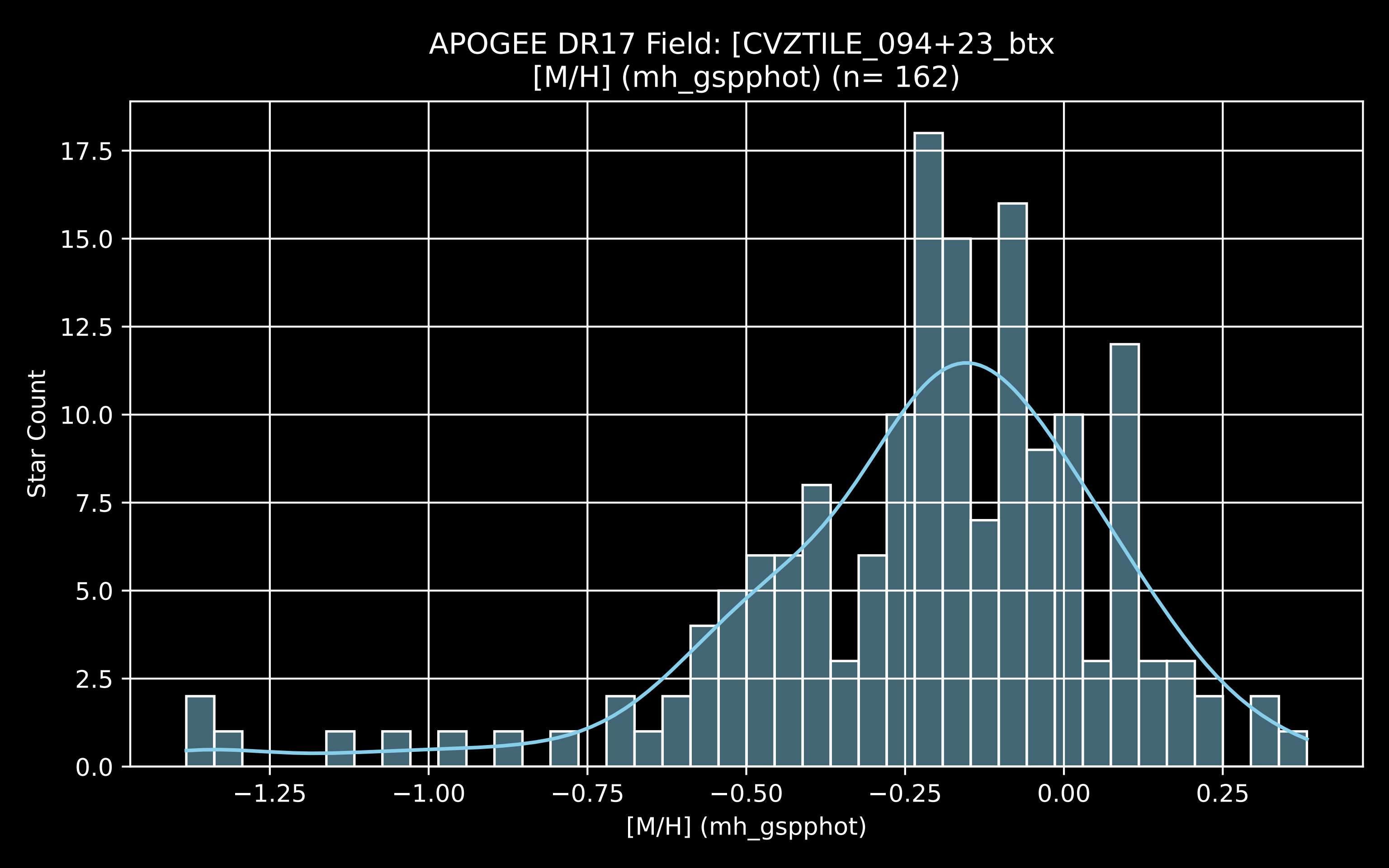

APOGEE DR17 Field [CVZTILE_094+23_btx]
Stars (n= 165)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−2
0
2
4
6
8
10
12
0.0
0.5
1.0
1.5
2.0
2.5

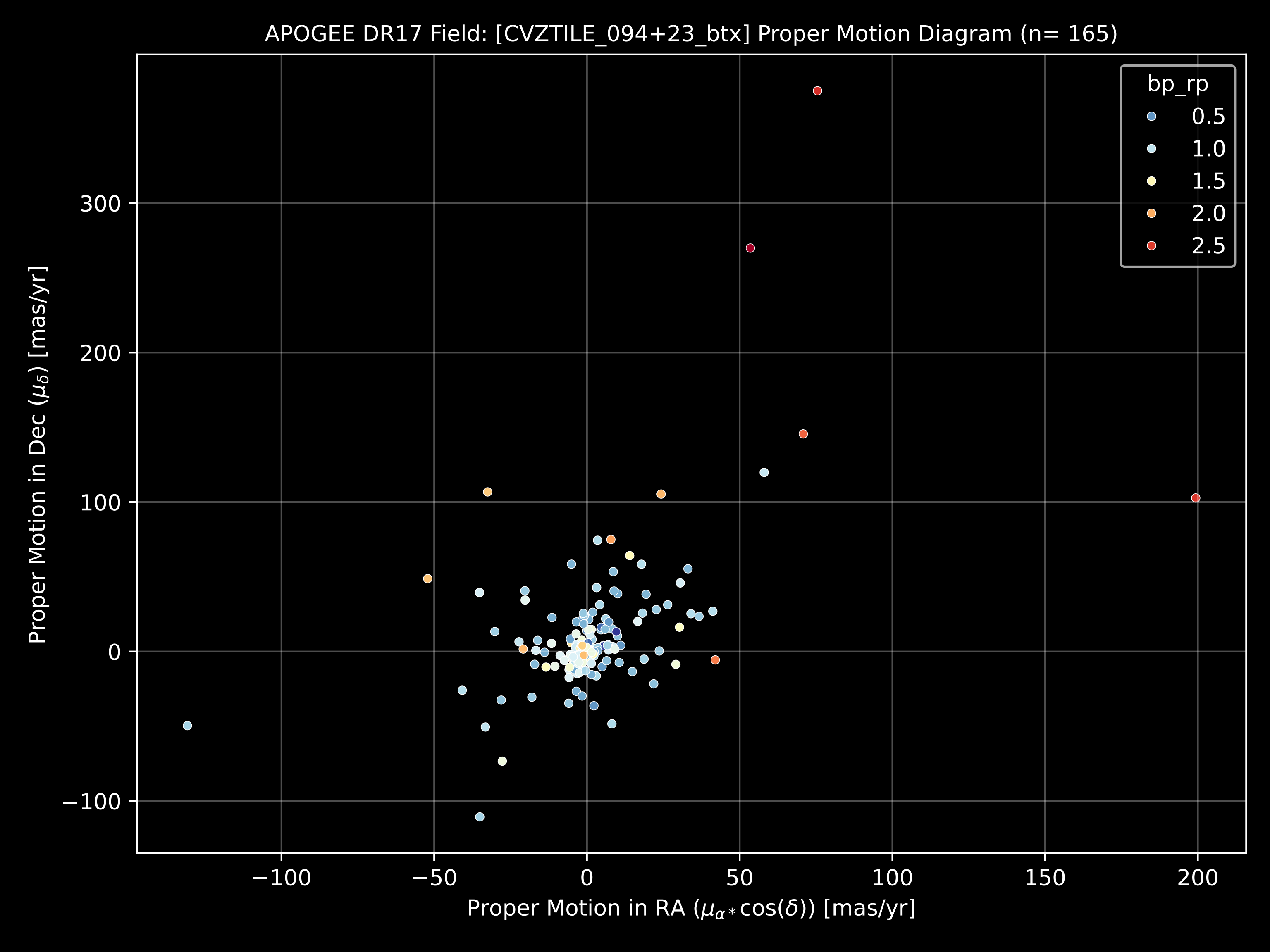

APOGEE DR17 Field: [CVZTILE_094+23_btx]
Stars: 165
Dec
Ra
bp_rp
64.5
64.0
63.5
63.0
62.5
62.0
61.5
22
23
24
25
26
27
28
2.5
2.0
1.5
1.0
0.5

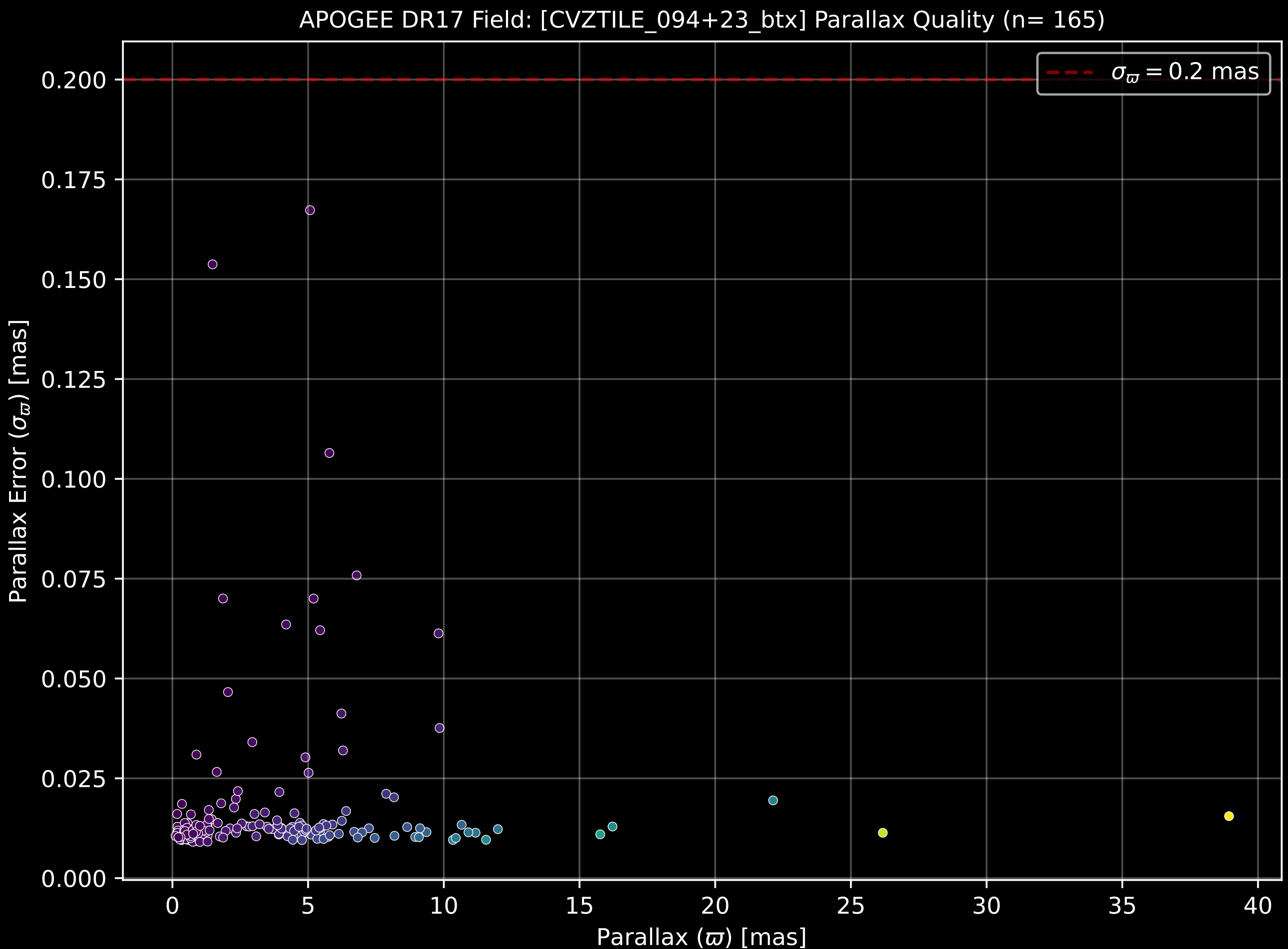


APOGEE DR17 Field: [CVZTILE_094+23_btx]
Stars: 165
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
64.5
64.0
63.5
63.0
62.5
62.0
61.5
22
23
24
25
26
27
28
2.5
2.0
1.5
1.0
0.5

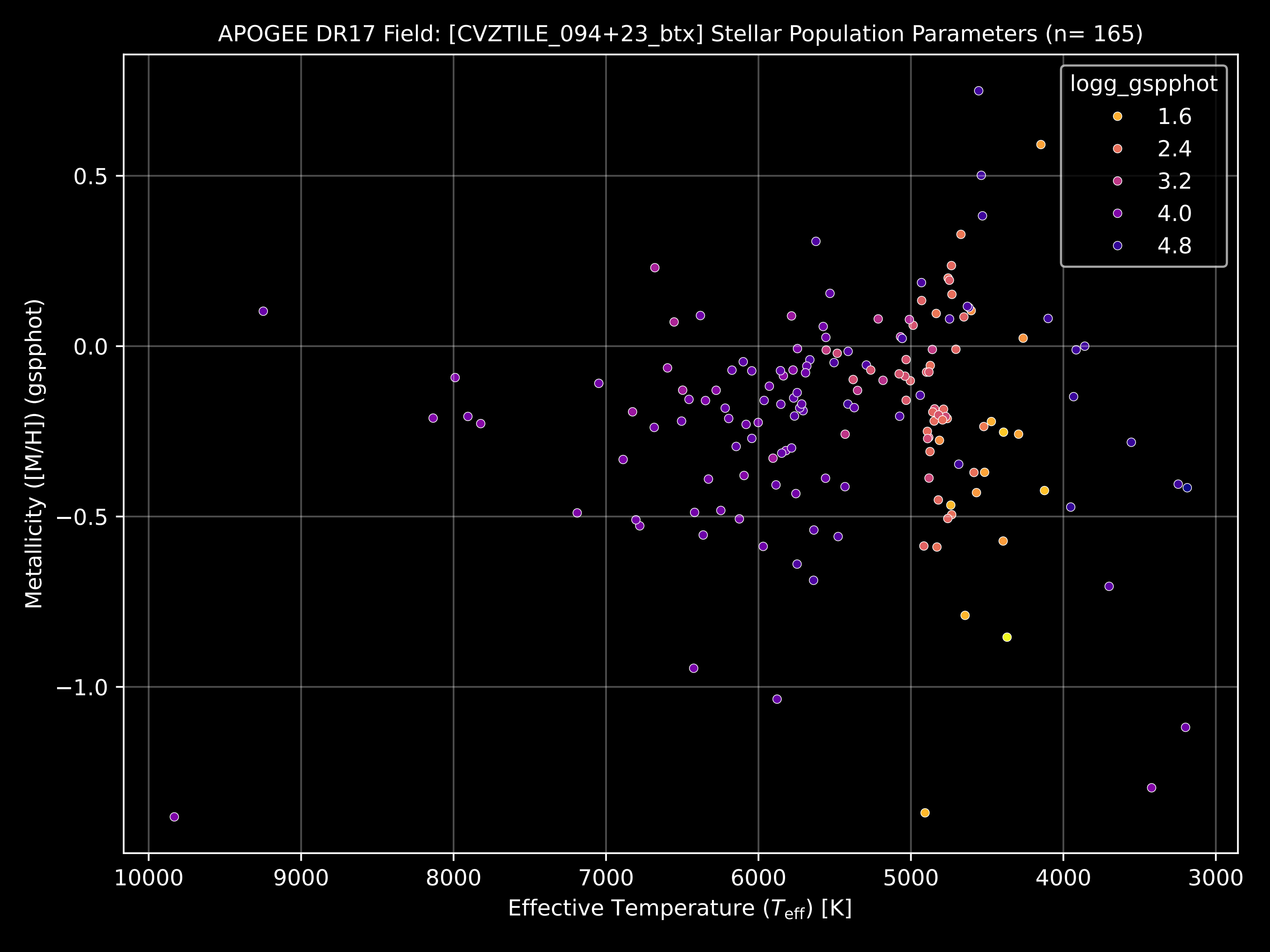

APOGEE DR17 Field: [CVZTILE_094+23_btx]
Count plot of Star Type
Count
0
10
20
30
40
50
60
turnoff
main_sequence
red_clump
red_giants
subgiants
Star Type

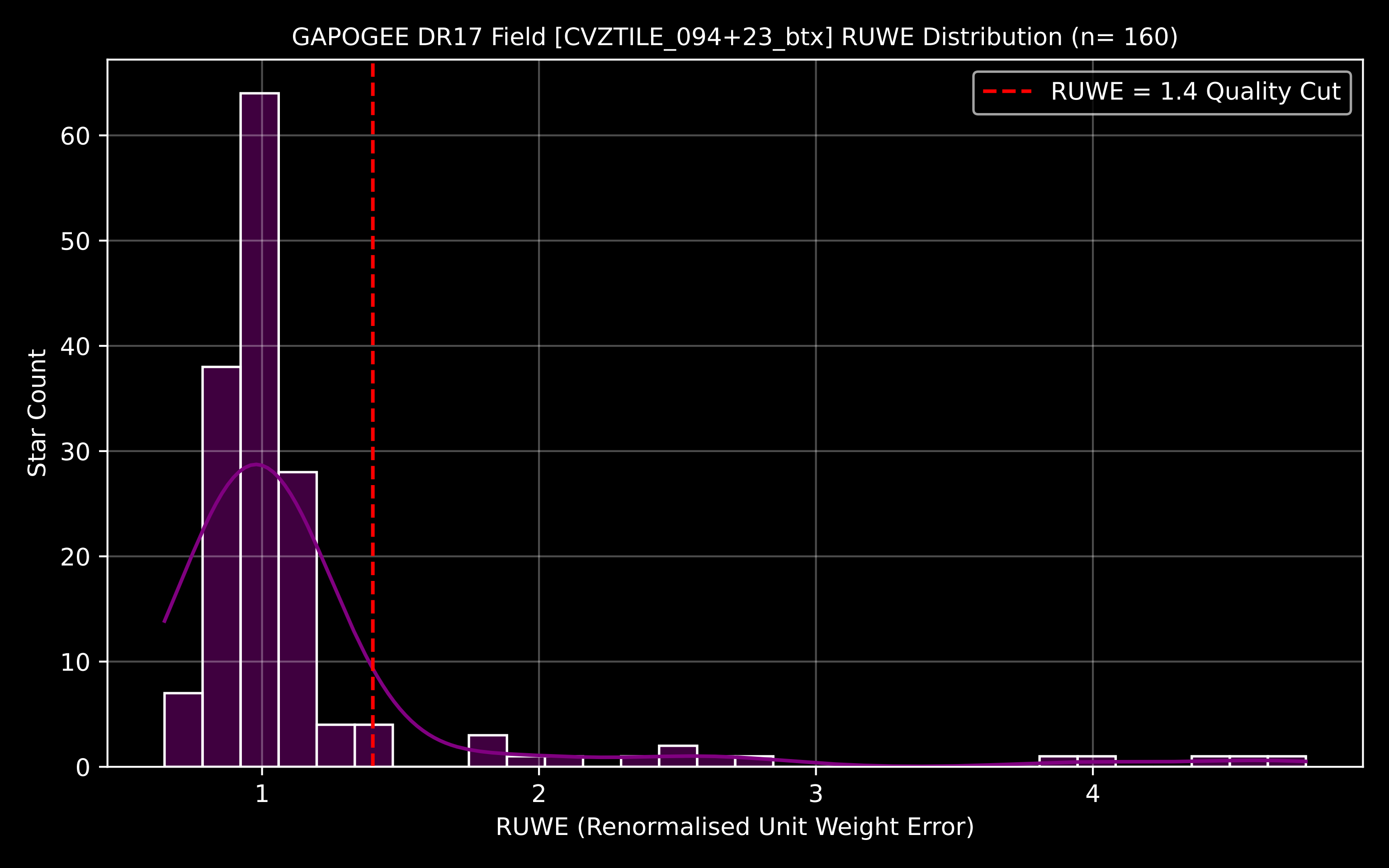

APOGEE DR17 Field: [CVZTILE_094+23_btx
[M/H] (mh_gspphot) (n= 162)
Star Count
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
[M/H] (mh_gspphot)

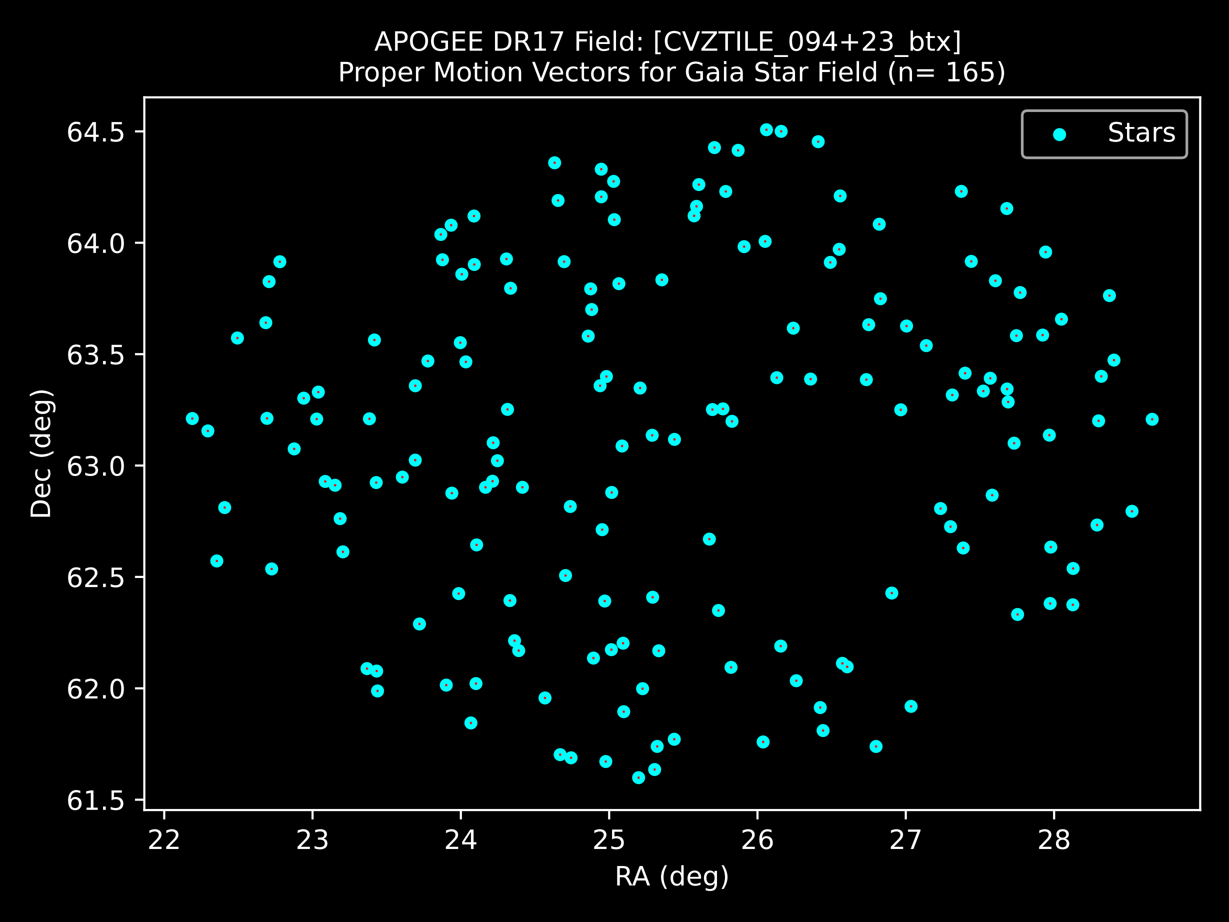

APOGEE DR17 Field: [CVZTILE_094+23_btx] Proper Motion Diagram (n= 165)
bp_rp
0.5
1.0
1.5
2.0
2.5
300
200
100
0
−100
−100
−50
0
50
100
150
200
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]



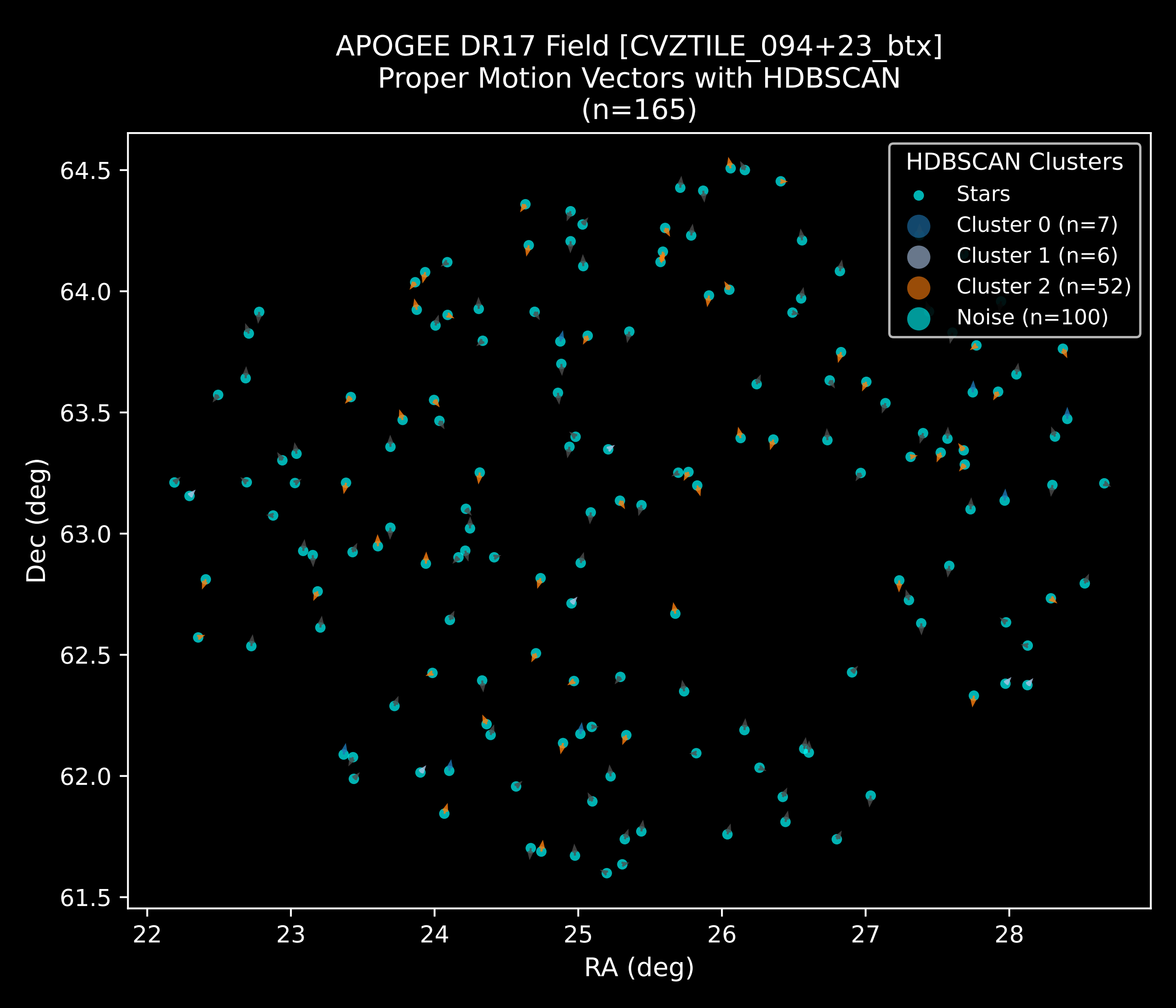
APOGEE DR17 Field: [CVZTILE_094+23_btx] Parallax Quality (n= 165)
$\sigma_{\varpi} = 0.2$ mas
Parallax Error (σ_{ϖ}) [mas]
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Parallax (ϖ) [mas]
0
5
10
15
20
25
30
35
40


APOGEE DR17 Field: [CVZTILE_094+23_btx] Stellar Population Parameters (n= 165)
Metallicity ([M/H]) (gspphot)
Effective Temperature (T_{eff}) [K]
logg_gspphot
1.6
2.4
3.2
4.0
4.8
0.5
0.0
−0.5
−1.0
10000
9000
8000
7000
6000
5000
4000
3000

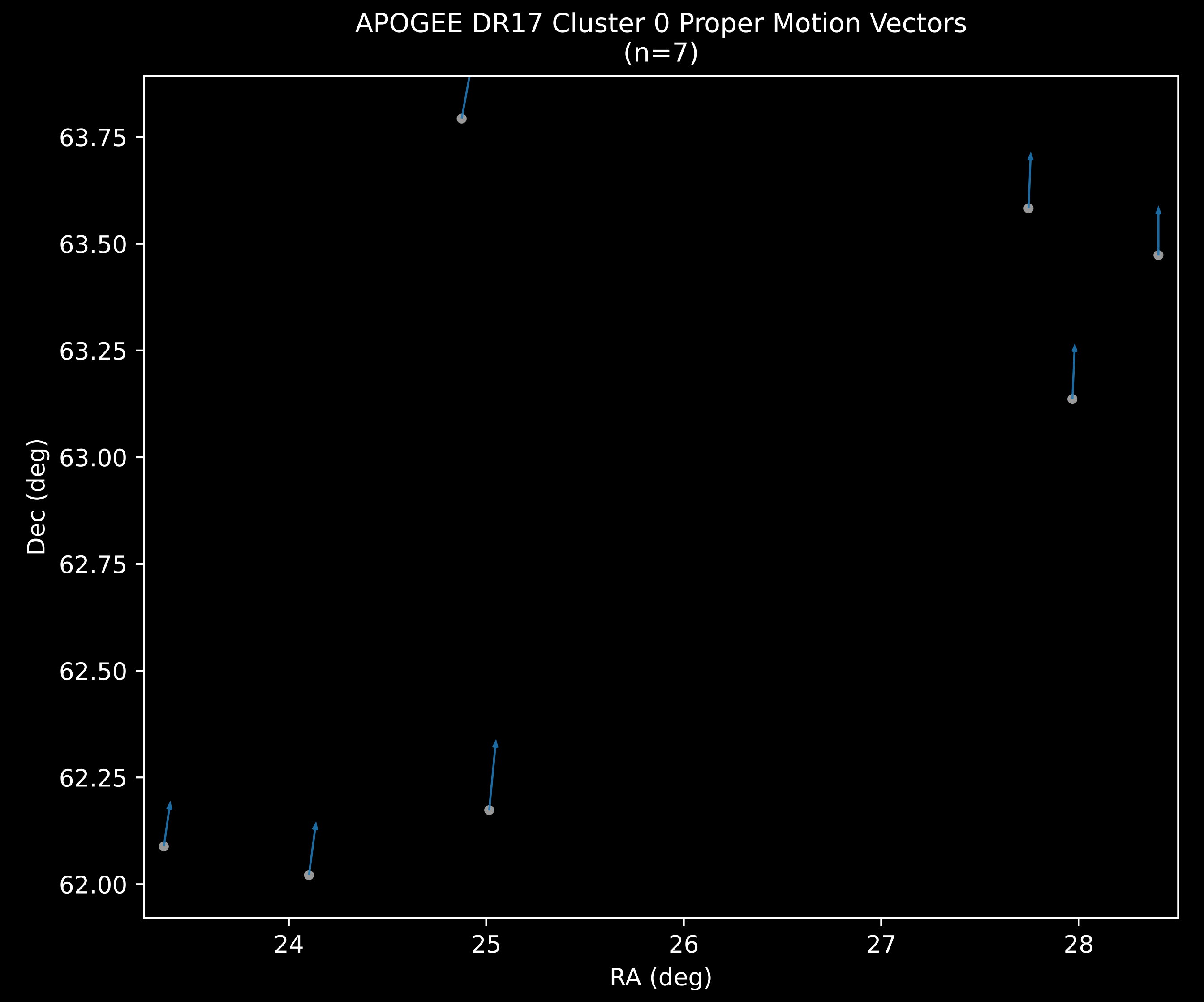

GAPOGEE DR17 Field [CVZTILE_094+23_btx] RUWE Distribution (n= 160)
RUWE = 1.4 Quality Cut
Star Count
0
10
20
30
40
50
60
1
2
3
4
RUWE (Renormalised Unit Weight Error)

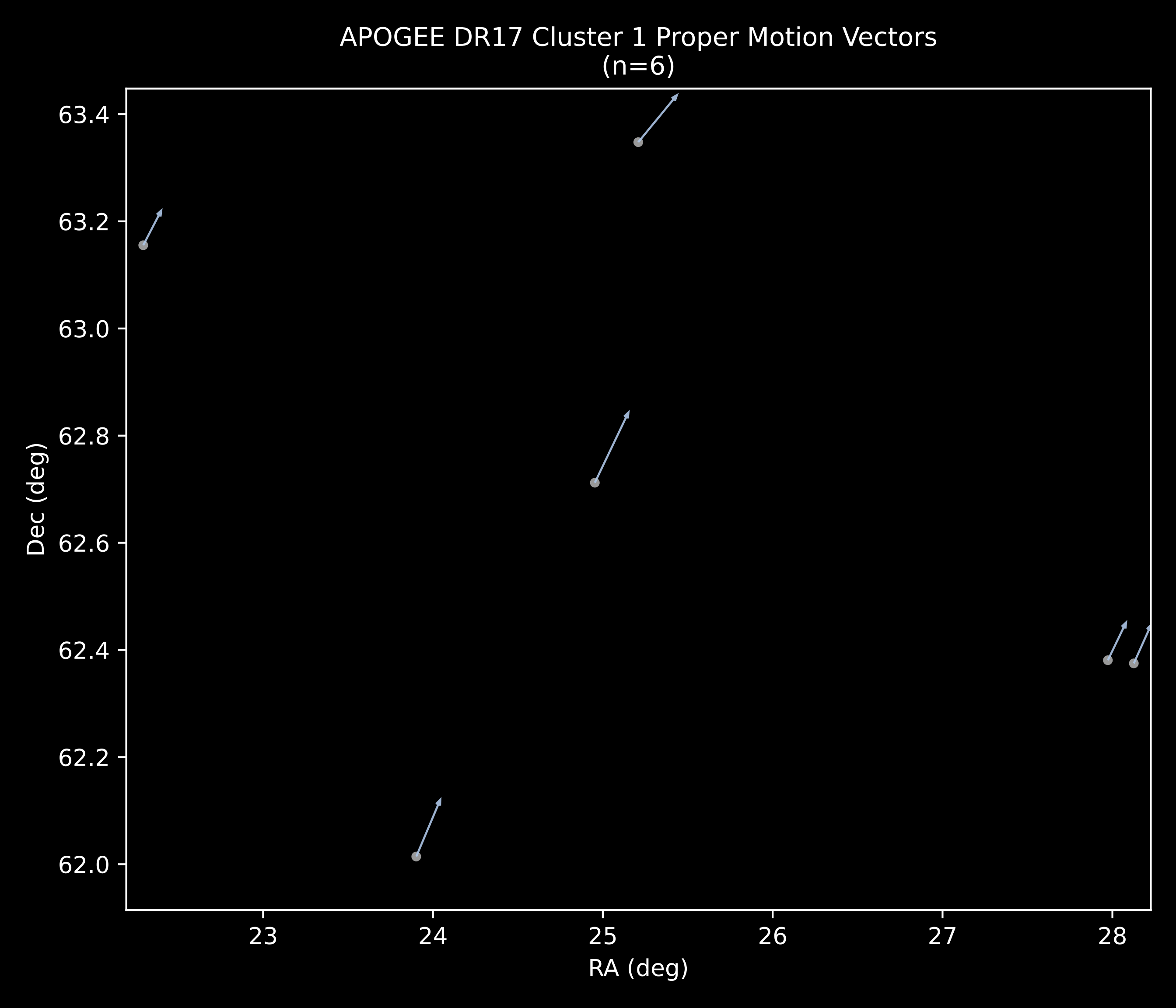


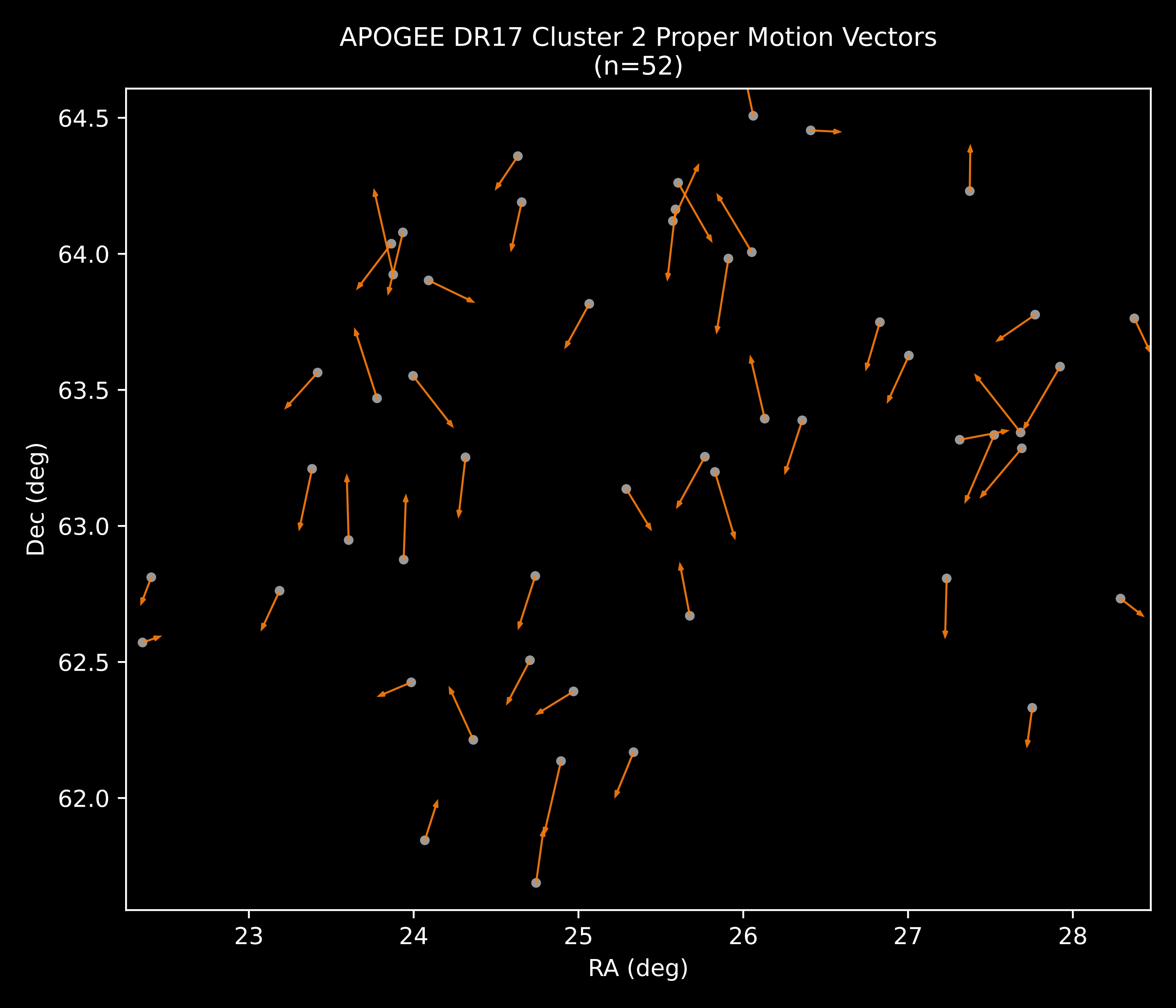
APOGEE DR17 Field: [CVZTILE_094+23_btx]
Proper Motion Vectors for Gaia Star Field (n= 165)
Stars
Dec (deg)
RA (deg)
64.5
64.0
63.5
63.0
62.5
62.0
61.5
22
23
24
25
26
27
28

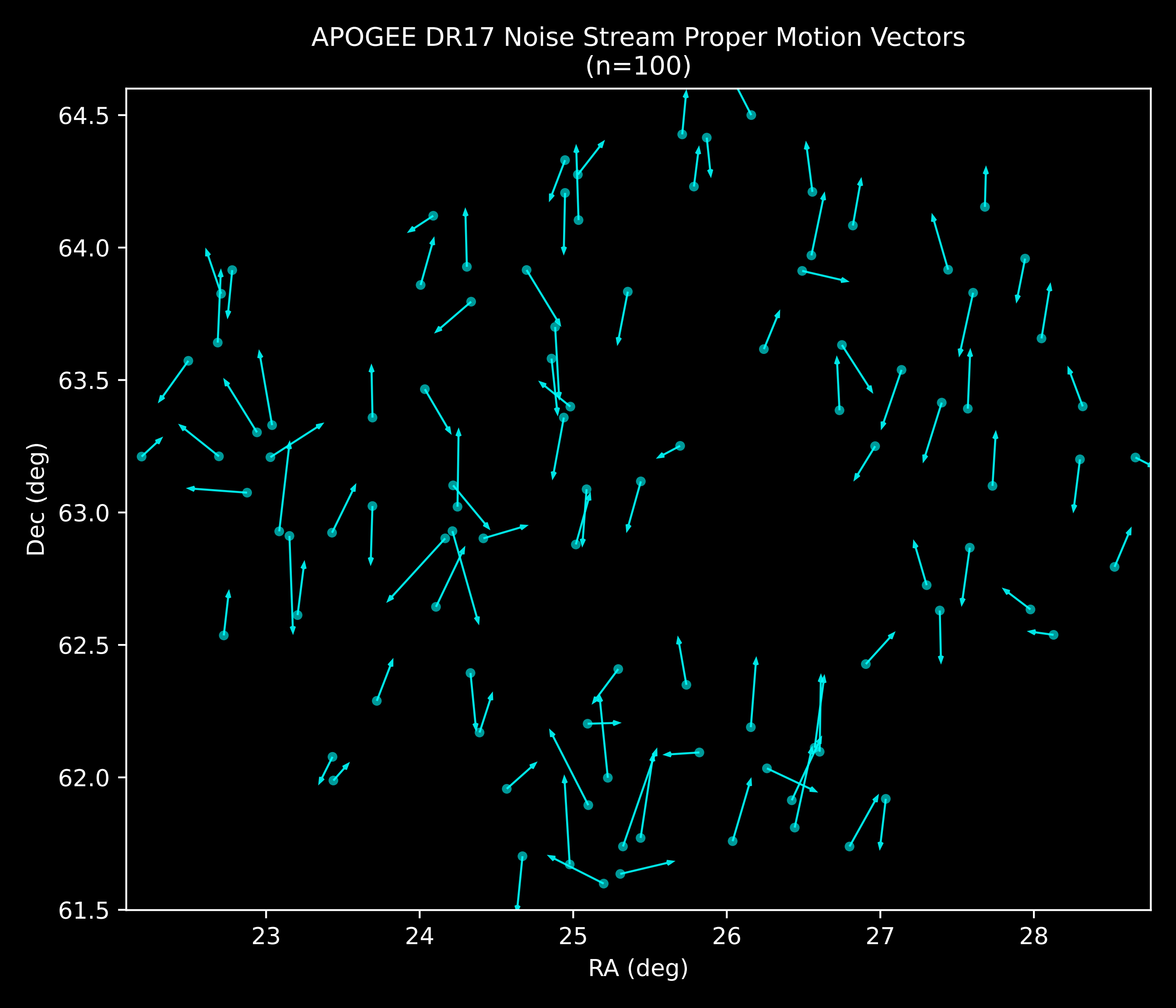


APOGEE DR17 Field [CVZTILE_094+23_btx]
Proper Motion Vectors with HDBSCAN
(n=165)
HDBSCAN Clusters
Stars
Cluster 0 (n=7)
Cluster 1 (n=6)
Cluster 2 (n=52)
Noise (n=100)
Dec (deg)
RA (deg)
64.5
64.0
63.5
63.0
62.5
62.0
61.5
22
23
24
25
26
27
28


APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=7)
Dec (deg)
63.75
63.50
63.25
63.00
62.75
62.50
62.25
62.00
RA (deg)
24
25
26
27
28


APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=6)
Dec (deg)
RA (deg)
63.4
63.2
63.0
62.8
62.6
62.4
62.2
62.0
23
24
25
26
27
28


APOGEE DR17 Cluster 2 Proper Motion Vectors
(n=52)
Dec (deg)
RA (deg)
64.5
64.0
63.5
63.0
62.5
62.0
23
24
25
26
27
28



APOGEE DR17 Noise Stream Proper Motion Vectors
(n=100)
64.5
64.0
63.5
63.0
62.5
62.0
61.5
Dec (deg)
23
24
25
26
27
28
RA (deg)